

"ABC Technologies — AI Customer Support System — Screenshots"

1) Screenshot 1 — RAG Pipeline Loading

```
● PS C:\Users\kumku\customer_support_ai> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& C:\Users\kumku\customer_support_ai\venv\Scripts\Activate.ps1)
(venv) PS C:\Users\kumku\customer_support_ai> cd C:\Users\kumku\customer_support_ai
(venv) PS C:\Users\kumku\customer_support_ai> venv\Scripts\activate
(venv) PS C:\Users\kumku\customer_support_ai> python main.py
[RAG] Initializing RAG pipeline...
[RAG] Loaded: documents/company_policy.txt
[RAG] Loaded: documents/pricing_guide.txt
[RAG] Loaded: documents/technical_manual.txt
[RAG] Loaded: documents/faq_document.txt
[RAG] Created 10 document chunks.
C:\Users\kumku\customer_support_ai\rag_pipeline.py:47: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecated in LangChain 0.2.2 and will be removed in 1.0. An updated version of the class exists in the `langchain-huggingface` package and should be used instead. To use it run `pip install -U `langchain-huggingface` and import as `from langchain_huggingface import HuggingFaceEmbeddings`.
  embeddings = HuggingFaceEmbeddings(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|█| 103/103 [00:00<00:00, 3879.
[RAG] Vector store created successfully.
[Memory] Database initialized successfully.
```

2) Screenshot 2 — Sales Agent Routing

```
🔥 Starting ABC Technologies AI Customer Support System...
✅ LangGraph workflow built successfully!

=====
👤 Customer: Alice (ID: CUST001)
❓ Query : What are the pricing plans available for your software?
=====

[Node] classify_intent running...
[Intent] Classified as: Sales

[Router] Routing to: Sales

[Node] retrieve_knowledge running...
[RAG] Retrieved 3 relevant chunks.

[Router] Routing to: Sales

[Node] sales_agent running...
[Sales Agent] Response generated.

[Node] supervisor_node running...
[Supervisor] Response reviewed and approved.

[Node] generate_final_response running...
[Memory] Conversation saved for customer: CUST001
[Final] Response ready and saved to memory.

-----
📁 Department : Sales
🔑 Intent : Sales
✅ Approval : not_required

💡 Final Response:
Customer Query: What are the pricing plans available for your software?
Department: Sales
Agent Response to Review:
Thank you for considering ABC Technologies for your software needs. I'd be happy to walk y
hrough our pricing plans.

We offer three subscription plans to cater to different business sizes and requirements. O
rtar Plan is perfect for small teams or individuals, offering up to 5 users, 10 GB stora
and basic reporting at just $29/month. If you need more advanced features, our Professiona
an is a great option, providing up to 25 users, 50 GB storage, priority support, and advan
reporting for $79/month.
```

● Final Response:

Customer Query: What are the pricing plans available for your software?

Department: Sales

Agent Response to Review:

Thank you for considering ABC Technologies for your software needs. I'd be happy to walk you through our pricing plans.

We offer three subscription plans to cater to different business sizes and requirements. Our Starter Plan is perfect for small teams or individuals, offering up to 5 users, 10 GB storage, and basic reporting at just \$29/month. If you need more advanced features, our Professional Plan is a great option, providing up to 25 users, 50 GB storage, priority support, and advanced reporting for \$79/month.

For larger enterprises or businesses with complex needs, our Enterprise Plan offers unlimited users, 500 GB storage, 24/7 dedicated support, custom reporting, and API access for \$199/month. We also offer a 20% discount for annual payments, making our plans even more cost-effective.

If you're unsure about which plan is right for you, I recommend taking advantage of our 14-day free trial. This will give you hands-on experience with our software and allow you to test all the features and functionality before committing to a plan.

Additionally, you can upgrade or downgrade your plan at any time from your Dashboard, so you can adjust your subscription as your business grows or changes.

To make it easier to compare our plans, here is a summary:

- **Starter Plan**: \$29/month (up to 5 users, 10 GB storage) - Annual: \$278/year (20% off)
- **Professional Plan**: \$79/month (up to 25 users, 50 GB storage) - Annual: \$758/year (20% off)
- **Enterprise Plan**: \$199/month (unlimited users, 500 GB storage) - Annual: \$1,910/year (20% off)

Which plan do you think best fits your needs? Or would you like to discuss further or explore our free trial option?

Approval Status: Approved

3) Screenshot 3 — Account Agent Routing

```
=====
👤 Customer: Bob (ID: CUST002)
❓ Query : I forgot my account password.
=====

[Node] classify_intent running...
[Intent] Classified as: Account

[Router] Routing to: Account

[Node] retrieve_knowledge running...
[RAG] Retrieved 3 relevant chunks.

[Router] Routing to: Account

[Node] account_agent running...
[Account Agent] Response generated.

[Node] supervisor_node running...
[Supervisor] Response reviewed and approved.

[Node] generate_final_response running...
[Memory] Conversation saved for customer: CUST002
[Final] Response ready and saved to memory.

=====
📁 Department : Account Management
🔍 Intent : Account
✅ Approval : not_required

● Final Response:
**Account Management Response: Password Reset Assistance**

Hello [Customer Name],

Thank you for reaching out to ABC Technologies' Account Management Support. I'm here to assist you with resetting your account password.

To reset your password, please follow these steps:

1. Go to the login page of your ABC Technologies account.
2. Click on the "Forgot Password" button.
3. Enter the email address associated with your account.
4. Check your inbox for an email from ABC Technologies with a password reset link. This link is valid for 24 hours, so please act quickly.
5. Click on the reset link and follow the prompts to create a new password.
```



```
Hello [Customer Name],

Thank you for reaching out to ABC Technologies' Account Management Support. I'm here to assist you with resetting your account password.

To reset your password, please follow these steps:

1. Go to the login page of your ABC Technologies account.
2. Click on the "Forgot Password" button.
3. Enter the email address associated with your account.
4. Check your inbox for an email from ABC Technologies with a password reset link. This link is valid for 24 hours, so please act quickly.
5. Click on the reset link and follow the prompts to create a new password.

If you don't receive the email, please check your spam folder, as it may have been filtered there. If you're still having trouble, please let me know and I'll be happy to assist you further.

To maintain the security of your account, we recommend using a strong and unique password, avoiding easily guessable information, and considering a password manager to generate and store complex passwords. Regularly updating your password is also essential for maintaining account security.

If you have any other questions or concerns, please don't hesitate to reach out. I'm here to help.

Best regards,
[Your Name]
Account Management Support Agent
ABC Technologies

**Approval Status:** Approved
```

4) Screenshot 4 — Technical Agent Routing

```
=====
👤 Customer: Carol (ID: CUST003)
❓ Query : My application crashes whenever I upload a file.
=====

[Node] classify_intent running...
[Intent] Classified as: Technical

[Router] Routing to: Technical

[Node] retrieve_knowledge running...
[RAG] Retrieved 3 relevant chunks.

[Router] Routing to: Technical

[Node] technical_agent running...
[Technical Agent] Response generated.

[Node] supervisor_node running...
[Supervisor] Response reviewed and approved.

[Node] generate_final_response running...
[Memory] Conversation saved for customer: CUST003
[Final] Response ready and saved to memory.

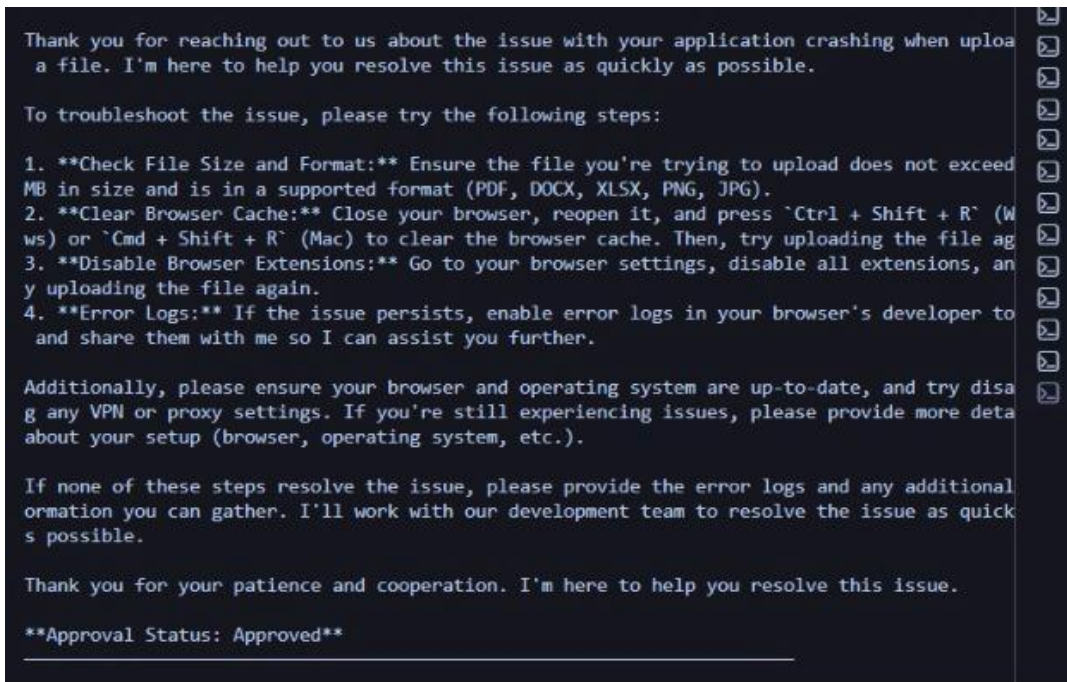
-----
📁 Department : Technical Support
🔍 Intent : Technical
✅ Approval : not_required

🗨️ Final Response:
**Final Response:**

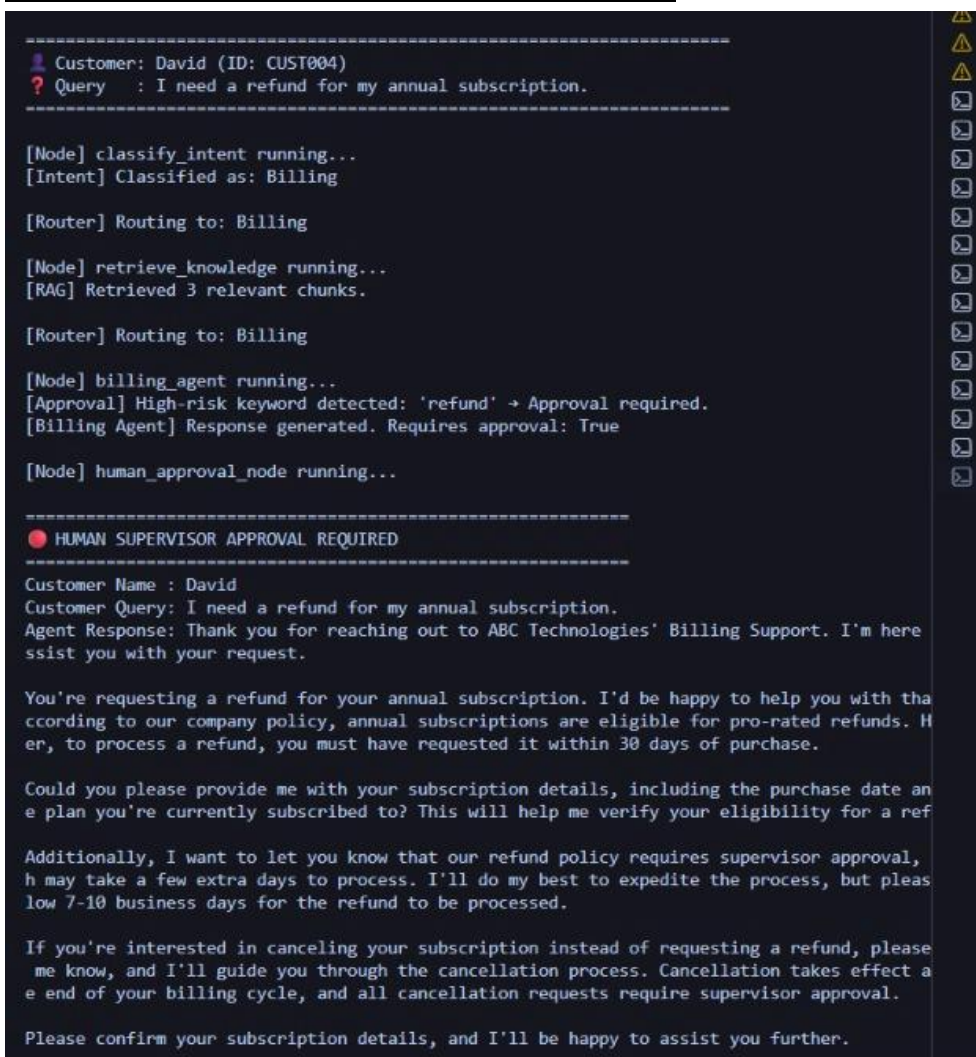
Thank you for reaching out to us about the issue with your application crashing when uploading a file. I'm here to help you resolve this issue as quickly as possible.

To troubleshoot the issue, please try the following steps:

1. **Check File Size and Format:** Ensure the file you're trying to upload does not exceed 10 MB in size and is in a supported format (PDF, DOCX, XLSX, PNG, JPG).
2. **Clear Browser Cache:** Close your browser, reopen it, and press 'Ctrl + Shift + R' (Windows) or 'Cmd + Shift + R' (Mac) to clear the browser cache. Then, try uploading the file again.
3. **Disable Browser Extensions:** Go to your browser settings, disable all extensions, and try uploading the file again.
4. **Error Logs:** If the issue persists, enable error logs in your browser's developer tools and share them with me so I can assist you further.
```



5) Screenshot 5 — Human Supervisor Approval



Please confirm your subscription details, and I'll be happy to assist you further.

****Customer Response Required****

Please provide your subscription details, and I'll proceed with your request.

[Supervisor] Reviewing request...

[Supervisor] Request APPROVED 

[Node] supervisor_node running...

[Supervisor] Response reviewed and approved.

[Node] generate_final_response running...

[Memory] Conversation saved for customer: CUST004

[Final] Response ready and saved to memory.

📁 Department : Billing
🔗 Intent : Billing
✅ Approval : approved

💡 Final Response:

****Final Response:****

Thank you for reaching out to ABC Technologies' Billing Support. I'm here to assist you with our request.

You're requesting a refund for your annual subscription. I'd be happy to help you with that. According to our company policy, annual subscriptions are eligible for pro-rated refunds. However, to process a refund, you must have requested it within 30 days of purchase.

Could you please provide me with your subscription details, including the purchase date and the plan you're currently subscribed to? This will help me verify your eligibility for a refund.

Additionally, I want to let you know that our refund policy requires supervisor approval, which may take a few extra days to process. I'll do my best to expedite the process, but please allow 7-10 business days for the refund to be processed.

If you're interested in canceling your subscription instead of requesting a refund, please let me know, and I'll guide you through the cancellation process. Cancellation takes effect at the end of your billing cycle, and all cancellation requests require supervisor approval.

Please confirm your subscription details, and I'll be happy to assist you further.

****Validation Notes:****

Additionally, I want to let you know that our refund policy requires supervisor approval, which may take a few extra days to process. I'll do my best to expedite the process, but please allow 7-10 business days for the refund to be processed.

If you're interested in canceling your subscription instead of requesting a refund, please let me know, and I'll guide you through the cancellation process. Cancellation takes effect at the end of your billing cycle, and all cancellation requests require supervisor approval.

Please confirm your subscription details, and I'll be happy to assist you further.

****Validation Notes:****

- The response is accurate and based on company policy.
- The tone is professional and empathetic.
- The response is complete and actionable.
- The information is free of errors or misleading information.

[Memory] Conversation saved for customer: CUST004

6) Screenshot 6 — Memory Recall

```
=====
👤 Customer: David (ID: CUST004)
❓ Query : What was my previous support issue?
=====

[Node] classify_intent running...
[Intent] Classified as: Account

[Router] Routing to: Account

[Node] retrieve_knowledge running...
[RAG] Retrieved 3 relevant chunks.

[Router] Routing to: Account

[Node] account_agent running...
[Account Agent] Response generated.

[Node] supervisor_node running...
[Supervisor] Response reviewed and approved.

[Node] generate_final_response running...
[Memory] Conversation saved for customer: CUST004
[Final] Response ready and saved to memory.

-----
📁 Department : Account Management
🔑 Intent : Account
✅ Approval : not_required

💡 Final Response:
Customer Query: What was my previous support issue?
Department: Account Management

Response:
Hello, I'm an Account Management Support Agent for ABC Technologies.

I'm happy to help with your current concern or any other issue you'd like to discuss. However, our system is designed to handle each support case independently, and we don't retain information about previous issues unless you've opened a new case related to it.

If you'd like to access your account information or review previous interactions with our support team, I can guide you on how to do so. Please let me know how I can assist you further.

(I made minor adjustments to improve the tone and clarity of the response, while maintaining its accuracy and professionalism required.)
```

	customer...	customer...	query	intent	response	timestamp	
	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	
1	1	CUST001	Alice	What are the pricing plans available for your software?	Sales	Customer Query: What are the pricing plans available fo...	2026-06-24 21:35:47
2	2	CUST002	Bob	I forgot my account password.	Account	**Account Management Response: Password Reset Assi...	2026-06-24 21:35:48
3	3	CUST003	Carol	My application crashes whenever I upload a file.	Technical	**Final Response** Thank you for reaching out to us ab...	2026-06-24 21:35:51
4	4	CUST004	David	I need a refund for my annual subscription.	Billing	The response is well-written and follows the required ga...	2026-06-24 21:35:52
5	5	CUST004	David	I need a refund for my annual subscription.	Billing	Your refund request has been submitted and approved ...	2026-06-24 21:35:53
6	6	CUST004	David	What was my previous support issue?	Account	Agent Response to Review: Hello, this is [Your Name] fro...	2026-06-24 21:36:02
7	7	CUST001	Alice	What are the pricing plans available for your software?	Sales	Customer Query: What are the pricing plans available fo...	2026-06-25 10:27:12
8	8	CUST002	Bob	I forgot my account password.	Account	**Account Management Response: Password Reset Assi...	2026-06-25 10:27:14
9	9	CUST003	Carol	My application crashes whenever I upload a file.	Technical	**Final Response** Thank you for reaching out to us ab...	2026-06-25 10:27:16
10	10	CUST004	David	I need a refund for my annual subscription.	Billing	**Final Response** Thank you for reaching out to ABC T...	2026-06-25 10:27:19
11	11	CUST004	David	I need a refund for my annual subscription.	Billing	Your refund request has been submitted and approved ...	2026-06-25 10:27:19
12	12	CUST004	David	What was my previous support issue?	Account	Customer Query: What was my previous support issue? ...	2026-06-25 10:27:29
13							

memory.db

Rows: 12

	#	customer...	customer...	query	intent	response	timestamp
		Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
1	1	CUST001	Alice	What are the pricing plans available for your software?	Sales	Customer Query: What are the pricing plans available fo...	2026-06-24 21:35:47
2	2	CUST002	Bob	I forgot my account password.	Account	**Account Management Response: Password Reset Assis...	2026-06-24 21:35:48
3	3	CUST003	Carol	My application crashes whenever I upload a file.	Technical	**Final Response:** Thank you for reaching out to us ab...	2026-06-24 21:35:51
4	4	CUST004	David	I need a refund for my annual subscription.	Billing	The response is well-written and follows the required gu...	2026-06-24 21:35:53
5	5	CUST004	David	I need a refund for my annual subscription.	Billing	Your refund request has been submitted and approved ...	2026-06-24 21:35:53
6	6	CUST004	David	What was my previous support issue?	Account	Agent Response to Review: Hello, this is [Your Name] fro...	2026-06-24 21:36:03
7	7	CUST001	Alice	What are the pricing plans available for your software?	Sales	Customer Query: What are the pricing plans available fo...	2026-06-25 10:27:12
8	8	CUST002	Bob	I forgot my account password.	Account	**Account Management Response: Password Reset Assis...	2026-06-25 10:27:14
9	9	CUST003	Carol	My application crashes whenever I upload a file.	Technical	**Final Response:** Thank you for reaching out to us ab...	2026-06-25 10:27:16
10	10	CUST004	David	I need a refund for my annual subscription.	Billing	**Final Response:** Thank you for reaching out to ABC T...	2026-06-25 10:27:19
11	11	CUST004	David	I need a refund for my annual subscription.	Billing	Your refund request has been submitted and approved ...	2026-06-25 10:27:19
12	12	CUST004	David	What was my previous support issue?	Account	Customer Query: What was my previous support issue? ...	2026-06-25 10:27:29

	customer...	customer...	query	intent	response	timestamp
	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
1	1	CUST001	Alice	What are the pricing plans available for your software?	Sales	Customer Query: What are the pricing plans available fo... 2026-06-24 21:35:47
2	2	CUST002	Bob	I forgot my account password.	Account	**Account Management Response: Password Reset Assis... 2026-06-24 21:35:48
3	3	CUST003	Carol	My application crashes whenever I upload a file.	Technical	**Final Response:** Thank you for reaching out to us ab... 2026-06-24 21:35:51
4	4	CUST004	David	I need a refund for my annual subscription.	Billing	The response is well-written and follows the required gu... 2026-06-24 21:35:52
5	5	CUST004	David	I need a refund for my annual subscription.	Billing	Your refund request has been submitted and approved ... 2026-06-24 21:35:53
6	6	CUST004	David	What was my previous support issue?	Account	Agent Response to Review: Hello, this is [Your Name] fro... 2026-06-24 21:36:02
7	7	CUST001	Alice	What are the pricing plans available for your software?	Sales	Customer Query: What are the pricing plans available fo... 2026-06-25 10:27:12
8	8	CUST002	Bob	I forgot my account password.	Account	**Account Management Response: Password Reset Assis... 2026-06-25 10:27:14
9	9	CUST003	Carol	My application crashes whenever I upload a file.	Technical	**Final Response:** Thank you for reaching out to us ab... 2026-06-25 10:27:16
10	10	CUST004	David	I need a refund for my annual subscription.	Billing	**Final Response:** Thank you for reaching out to ABC T... 2026-06-25 10:27:19
11	11	CUST004	David	I need a refund for my annual subscription.	Billing	Your refund request has been submitted and approved ... 2026-06-25 10:27:19
12	12	CUST004	David	What was my previous support issue?	Account	Customer Query: What was my previous support issue? ... 2026-06-25 10:27:29

7) Screenshot 7 — All Queries Completed

```

✅ All demonstration queries completed successfully!
❖ (venv) PS C:\Users\kumku\customer_support_ai>

```